



Lecture 5



Disinfection and Sterilization

Dr. Rania Hamed
Lecturer of Microbiology

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Intended Learning Outcomes (ILOs)

Understand sterilization and
Disinfection processes

Distinguish between antiseptics and
disinfectants

Categorize equipment by infection
risk level

Select appropriate decontamination
methods

Apply correct infection control measures in
nursing practice

What is Decontamination?

- ❑ Decontamination is the use of **physical or chemical** methods to **remove, inactivate, or** destroy **pathogens on objects, making them safe for handling** by healthcare personnel and patients.
- ❑ This critical infection control procedure includes **three sequential steps** that must be performed in order to **ensure safety**.



Three Essential Steps of Decontamination



Step 1: Cleaning

Physical removal of soil
and organic matter



Step 2: Disinfection

Destruction of most
pathogenic
microorganisms



Step 3: Sterilization

Complete destruction
of all microorganisms

Step 1: Cleaning

- It is the process that Removes foreign material including soil, organic matter, and microorganisms from object surfaces
- Cleaning must always be performed **before disinfection or sterilization**

Organic debris can shield microorganisms
and prevent effective disinfection



Step 2: Disinfection

Definition

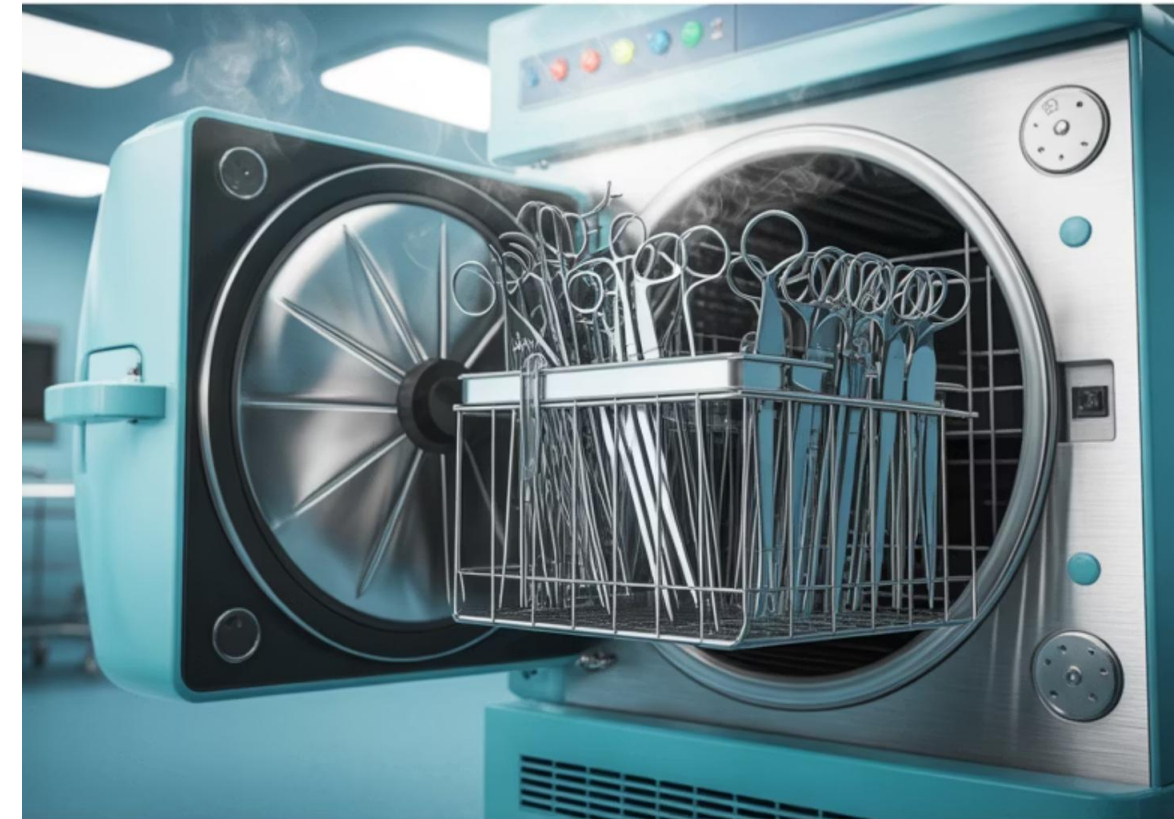
- It is The destruction of **most** pathogenic microorganisms on inanimate objects, but not necessarily bacterial spores.
- Spores can survive disinfection processes and require sterilization for complete elimination.



Step 3: Sterilization

❏ Complete Microbial Destruction

Sterilization is the only process that destroys all forms of microorganisms, including vegetative bacteria, fungi, viruses, and highly resistant bacterial spores.



Methods of Sterilization

Physical Methods

- Heat (moist and dry)
- Filtration
- Radiation

Chemical Methods

- Liquid sterilants
- Gas sterilants
- Plasma systems

Physical Methods of Sterilization

1-Heat Sterilization

Moist heat (autoclave) and
dry heat (hot air oven)

2-Filtration

Mechanical removal of
microorganisms from fluids

3-Radiation

Ionizing radiation for
disposable medical items

1- Heat sterilization

HEAT above 100° C

AUTOCLAVE



**HOT AIR
OVEN**



a-Moist Heat Sterilization: The Autoclave

Working Principle

Steam under pressure raises water's boiling point above 100°C, similar to a pressure cooker. This allows higher temperatures that effectively kill all microorganisms.

Structure

- Vertical metal cylinder with tight lid
- Safety valve and pressure gauge
- Manually operated tap



How the Autoclave Works

1-Pressure Increase

When pressure rises above atmospheric level, water's boiling point increases

2- Steam Penetration

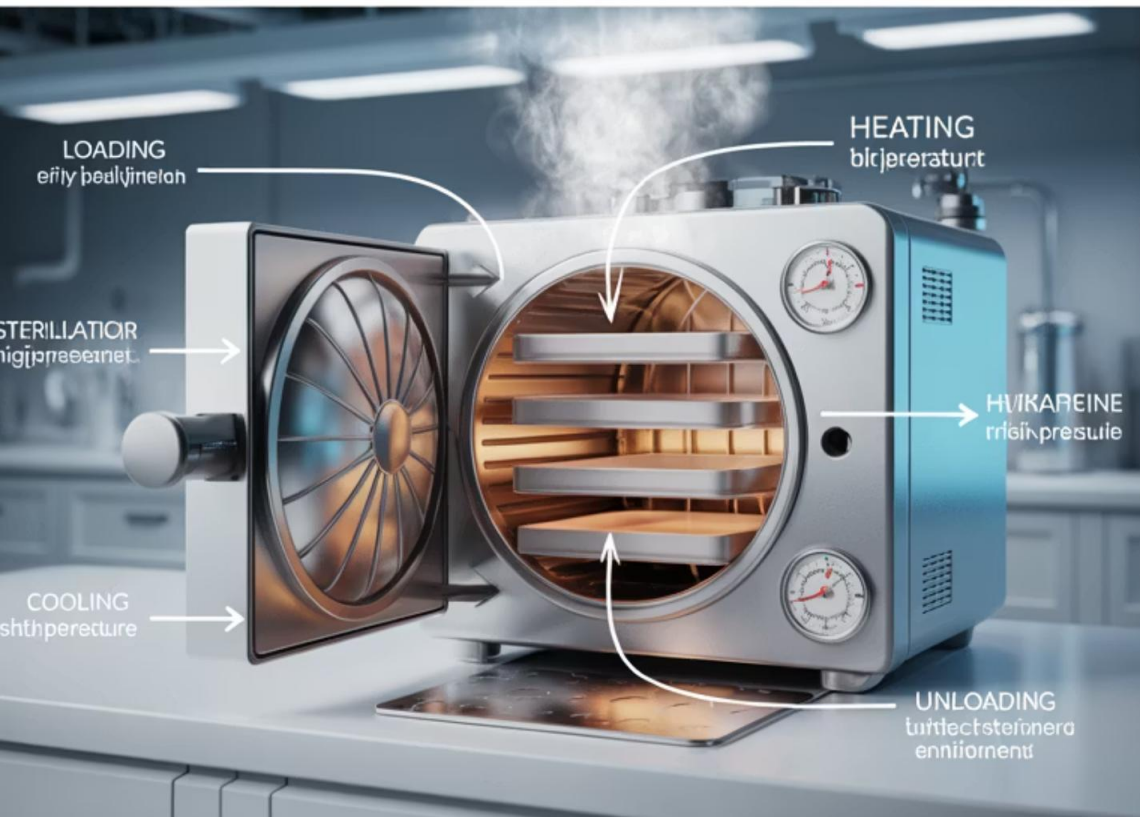
High-temperature steam penetrates materials and packages effectively

3- Latent Heat Release

Condensing steam releases latent heat energy directly into microorganisms

4-Complete Kill

Combined factors destroy all microbial life including resistant spores



Timing of use

Standard Cycle

121°C for minimum 15 minutes

Used for **routine** instrument sterilization

Flash Cycle

134°C for 3 minutes

Used for **urgent** surgical instruments

Uses: Autoclave Sterilization of

1. Surgical Instruments

Metal tools (scissors- forceps)

2. Textiles and Fabrics

Surgical dressings, cotton, gauze, gowns, and bed linens

3. Culture Media

Laboratory media (nutrient media, blood agar)
not destroyed by heat exposure



B- Dry Heat Sterilization: Hot Air Oven

Working Principle

Dry heat kills microorganisms through oxidative damage and charring of cellular components, requiring **higher temperatures** and **longer exposure times** than moist heat.



Hot Air Oven: Applications and Parameters



Uses: Sterilization of:

Glassware including **test tubes, Petri dishes, flasks and pipettes.**



Operating Conditions

Temperature: **170–180°C**

Duration: **1–2 hours**

Dry heat is preferred for items that might be damaged by moisture or need to remain completely dry.

Quiz: Test Your Knowledge

Question 1



Sterilization is a term used to describe:

1. Killing vegetative forms only
2. Reduction to a safe level
3. Destruction of all forms of organisms including spores
4. Swabbing skin before injection

Answer: Question 1

Option C ✓

Sterilization is the destruction of all forms of microorganisms including spores. This is what distinguishes it from disinfection, which does not eliminate spores.

2- Filtration Sterilization

Filtration is used for **heat-sensitive fluids** that would be destroyed by autoclaving. The process mechanically removes microorganisms by passing **liquids** through membranes with defined **pore sizes**.

Uses:

- Serum and plasma products
- Culture media with heat-labile components
- Vaccines and biological products
- Antibiotics and hormones



3- Radiation Sterilization

Ionizing Radiation also called **Cold sterilization**

e.g: Gamma rays

These have a high penetrating power and no residual radioactivity on products.

Uses:

Sterilization of disposable plastics: syringes, catheters, gloves, IV tubing, and sealed surgical packs.

Chemical Sterilization Methods

Used for **heat-sensitive** or **moisture-sensitive** critical items that cannot withstand autoclaving.

Mainly used for heat labile critical items
(e.g. endoscopes)



Quiz

Question 3

True or False:

If instruments will be sterilized, cleaning with (detergent and Water) is not necessary.

Answer: Question 3

FALSE ✓

Cleaning must always precede disinfection or sterilization. Organic soil and debris can shield microorganisms and prevent the sterilization process from working effectively. This is a critical principle in infection control.

Disinfection



Definition

Disinfection destroys most pathogenic microorganisms—bacteria, viruses, and fungi—but does not eliminate bacterial spores.

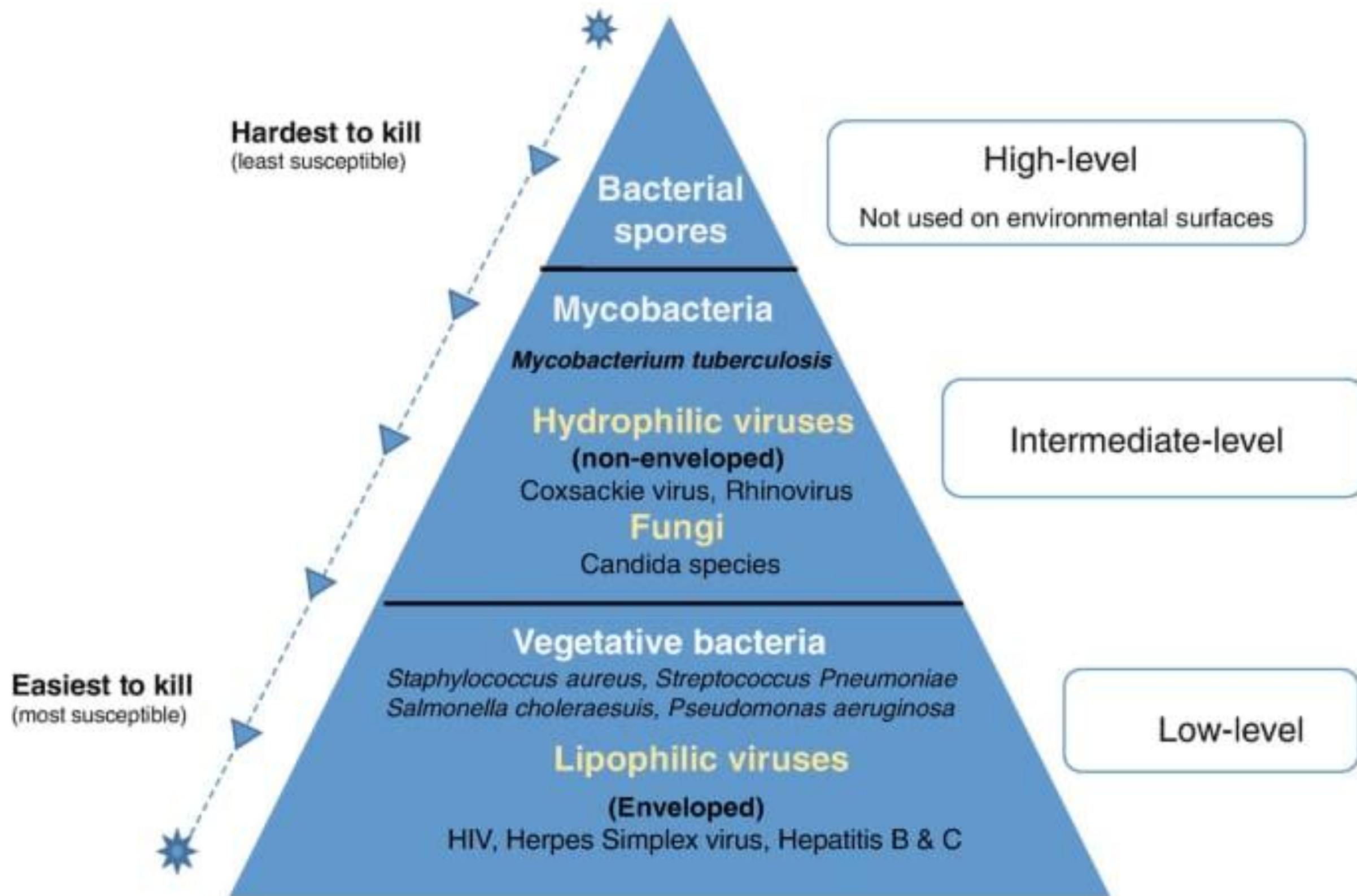
Three Levels

- 1. High-level: Kills all except large numbers of spores**
- 2. Intermediate-level: Kills Mycobacteria, most viruses, fungi**
- 3. Low-level: Kills some viruses and bacteria only**

The choice of the method:

The choice of the method (cleaning, disinfection or sterilization) depends on a number of factors:

- ◆ **Material of object.**
- ◆ **Number and type of organisms involved.**
- ◆ **Risk of infection to patients or staff.**





The Spaulding Classification System

This evidence-based system categorizes **medical items** by **infection risk** based on their **intended use and tissue contact**. It guides selection of appropriate reprocessing methods.

Spaulding Classification Overview

Critical (High Risk)

Sterile tissues/bloodstream → Sterilization required

Semi-Critical (Intermediate)

Mucous membranes → High-level disinfection

Non-Critical (Low Risk)

Intact skin → Cleaning or low-level disinfection

Spaulding Classification for Disinfection

Patient Contact	Device Classification	Disinfection Level or Sterilization	Examples	
Intact skin or non-patient contacting	Non-critical	Low-Level Disinfection or Intermediate-Level Disinfection		
Mucous membranes or non-intact skin	Semi-critical	High-Level Disinfection		
Sterile areas of the body, including blood pathway contact	Critical	Sterilization		

Non-Critical Items (Low Risk)

Definition

Items that **contact intact skin** but not mucous membranes. Intact skin serves as an effective barrier to most microorganisms.

Examples

- Blood pressure cuffs
- Stethoscopes
- furniture and tables
- Floors, walls, and sinks



Required Process

Cleaning or low-level disinfection

(e.g., **alcohol wipes**) is sufficient for these items.

Semi-Critical Items (Intermediate Risk)

Endoscopes

Contact mucous membranes of GI or respiratory tract

Laryngoscopes

Used for airway visualization and intubation

Thermometers

Oral and rectal thermometers require HLD

Vaginal Instruments

Contact mucous membranes during examination

Required Process

- All require cleaning + high-level disinfection (boiling or chemical agents like glutaraldehyde).

Critical Items (High Risk)

Surgical and Dental Instruments

Enter sterile tissues or vascular system

Catheters and Implants

Placed directly into bloodstream or sterile body cavities

Endoscope (laparoscope , arthroscope)

Penetrates through sterile peritoneal cavity

Required Process

All require cleaning + sterilization



Disinfectant vs. Antiseptic

Disinfectants

Function: Bactericidal (kills microorganisms)

Use: Inanimate objects only—
toxic to living tissues

Examples: Phenol, chlorine, bleach

Antiseptics

Function: Bacteriostatic (inhibits growth)

Use: Living tissues (wounds)—less toxic formulations

Examples: Iodine compounds, alcohol,
chlorhexidine

Note: Some chemicals like (alcohol) can function as both,
depending on concentration and application.

Common Antiseptics (Tissues)



Alcohol (70%)

Used as swabs, sprays, hand rubs at 70% concentration.
Must contain glycerin to prevent skin dryness. Avoid on wounds or mucous membranes.



Chlorhexidine

Ideal for **mouthwash** and skin antiseptic. Prolonged use may cause tooth discoloration over time.



Iodine Compounds

Betadine is best for **mucous membranes** and **open wound dressings**. Provides broad-spectrum antimicrobial activity.



Hydrogen Peroxide (3–6%)

Used for **wound antiseptics** and **mouth gargling**. Effervescent action helps clean debris from wounds.

Remember: Phenol is NOT an antiseptic—too toxic for skin.

Dettol should not be used for hand or wound cleaning.



Managing Blood and Body Fluid Spills



Step 1: Protection

Put on clean disposable gloves immediately



Step 2: Remove Soil

Use absorbent towels to remove visible organic material



Step 3: Apply Disinfectant

Apply **chlorine solution** (bleach) for **15 minutes** contact time



Step 4: Wipe Clean

Wipe chlorine from outside to inside using clean towels



Step 5: Final Cleaning

Clean area with soap and water, then dry with clean towels

In this short video, you will learn Management of Blood and Body Fluid Spills in Healthcare Settings





Quiz Time:

Which of the following can be sterilized by filtration?

- A.** Culture media with heat-labile components
- B.** Surgical instruments
- C.** Bed linens

Moist heat sterilization is achieved using _____?

- | | |
|------------------------|----------------------|
| A. Hot air oven | B. Autoclave |
| C. Radiation | D. Filtration |

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Quiz Time:

Disinfection is defined as:

- A.** Destruction of all microorganisms including spores
- B.** Destruction of most pathogenic microorganisms but not spores
- C.** Removal of organic matter only
- D.** Killing spores only

Decontamination *refers to*:

- A.** Complete destruction of microorganisms
- B.** Removal or inactivation of pathogens from objects to make safe
- C.** Killing bacterial spores only
- D.** Sterilization of surgical instruments

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Quiz Time:

The standard autoclave cycle for routine sterilization is:

- A. 100°C for 30 minutes**
- B. 121°C for 15 minutes**
- C. 134°C for 1 minute**
- D. 90°C for 20 minutes**

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Thank You



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